

Conditional Flow Matching vs. Diffusion Models: A systematic comparison on CIFAR-10

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Abstract

We compare Optimal Transport Conditional Flow Matching (OT-CFM) against a DDPM trained from scratch on CIFAR-10 and DDIM sampling from the same DDPM checkpoint, using a shared UNet backbone. OT-CFM achieves a best FID of 10.65 with RK4 at 100 NFE, surpassing DDPM (FID 13.06 at 1000 steps) with 20× fewer function evaluations. We extend CFM with class-conditional generation using classifier-free guidance, and show that the deterministic CFM ODE supports inversion, reconstruction, and simple image editing on real CIFAR-10 images. Flow matching strictly dominates diffusion at every NFE budget tested.

1 Introduction

Diffusion models (3) achieve excellent image quality but require ~ 1000 denoising steps. DDIM (8) enables fewer-step sampling but quality saturates quickly. Flow matching (4, 5) learns a velocity field transporting noise to data along simple paths. With optimal transport coupling (9), OT-CFM produces straight trajectories easily integrated by ODE solvers in far fewer steps.

Contributions. (1) We train DDPM, OT-CFM, and class-conditional CFM from scratch on CIFAR-10, and evaluate DDIM as deterministic sampling from the same trained DDPM under a shared architecture; (2) we extend CFM to class-conditional generation with CFG (2) and demonstrate ODE inversion for reconstruction and image editing; (3) we systematically analyze ODE solvers across NFE budgets; (4) we show flow matching dominates diffusion at every NFE on CIFAR-10.

2 Background

DDPM. The forward process is:

$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$. A network ϵ_θ predicts the noise via $\mathcal{L} = \mathbb{E}\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2$, and sampling denoises over $T=1000$ steps (3).

DDIM (8) reinterprets the reverse process as a deterministic ODE, enabling fewer-step sampling from a DDPM-trained model.

CFM. Flow matching (4) learns $v_\theta(\mathbf{x}_t, t)$ generating a path from p_0 to p_1 . With OT interpolation (9): $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$, target $\mathbf{u}_t = \mathbf{x}_1 - \mathbf{x}_0$, loss $\mathcal{L}_{\text{CFM}} = \mathbb{E}\|v_\theta(\mathbf{x}_t, t) - \mathbf{u}_t\|^2$. Sampling solves the ODE $d\mathbf{x}/dt = v_\theta(\mathbf{x}_t, t)$ from $t=0$ to 1.

Solvers. Euler (1 NFE/step), Midpoint (2 NFE/step), RK4 (4 NFE/step).

3 Methods

3.1 Shared Architecture

The shared backbone used for DDPM and OT-CFM, and for DDIM sampling from the trained DDPM checkpoint, is a UNet (6): 128 base channels, multipliers [1, 2, 2, 2], 4 attention heads, GroupNorm-32, sinusoidal time embeddings. The

class-conditional variant uses a simplified UNet-style backbone with 128 base channels and an effective progression of 128→256→512 channels. It uses lightweight single-head self-attention at lower resolutions, and classifier-free guidance is enabled by replacing the class label with a null label for 10% of training examples, so that the model learns both conditional and unconditional predictions. The encoder follows a [1, 2, 2, 2] channel multiplier progression from a base of 128 channels (128→256→256→256), with strided convolutions for down-sampling and nearest-neighbor upsampling with convolutions in the decoder. Skip connections concatenate encoder features to corresponding decoder layers.

Time conditioning uses sinusoidal positional embeddings projected through a two-layer MLP into a scale-and-shift modulation applied to each residual block. Each block consists of two 3×3 convolutions with GroupNorm-32 normalization, SiLU activations, and dropout ($p=0.1$). Multi-head self-attention (4 heads) is applied at the 16×16 and 8×8 resolutions to capture long-range spatial dependencies, which is critical at CIFAR-10’s 32×32 resolution where the receptive field of convolutions alone is insufficient.

Why this architecture? The design follows the standard DDPM UNet from Ho et al. (3), which has become the de facto backbone for comparing generative frameworks on CIFAR-10. Using the same architecture for both DDPM and OT-CFM isolates the effect of the generative objective itself. The 128 base channel width balances capacity with training efficiency on a single GPU.

3.2 Class-Conditional Variant

The class-conditional CFM uses a simplified UNet with an effective channel progression of 128→256→512, providing higher capacity at the deepest resolution to capture class-specific features. Class information is injected through a learned embedding table (10 classes + 1 null token, projected to 256 dimensions), which modulates residual blocks in parallel with the time embedding via additive bias.

Classifier-free guidance (CFG) is enabled by replacing the

true class label with the null token for 10% of training samples. This trains the model to produce both conditional predictions $v_\theta(x_t, t, y)$ and unconditional predictions $v_\theta(x_t, t, \emptyset)$. At inference, the guided velocity is:

$$v_{\text{guided}} = v_\theta(x_t, t, \emptyset) + w \cdot (v_\theta(x_t, t, y) - v_\theta(x_t, t, \emptyset)) \quad (1)$$

where w is the guidance scale. Setting $w=1$ recovers the standard conditional prediction; $w>1$ amplifies the class-specific component of the velocity, improving per-class fidelity at the cost of sample diversity.

Self-attention in this variant uses single-head attention with explicit query-key-value projections and GroupNorm-8, trading multi-head expressiveness for reduced memory footprint given the larger channel dimensions.

3.3 Training Objectives

DDPM. The model is trained to predict the noise ϵ added at a random timestep $t \sim \mathcal{U}\{1, \dots, T\}$ with $T=1000$:

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon - \epsilon_\theta(x_t, t)\|^2], \quad x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad (2)$$

We use a cosine noise schedule ($\bar{\alpha}_t$ follows a shifted cosine curve), which provides more uniform signal-to-noise progression compared to the linear schedule and avoids wasting capacity on nearly noiseless or fully destroyed inputs.

OT-CFM. The model learns the velocity field that transports noise to data along the optimal-transport conditional path (4):

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{x_0 \sim \mathcal{N}(0, I), x_1 \sim p_{\text{data}}, t \sim \mathcal{U}(0, 1)} [\|v_\theta(x_t, t) - (x_1 - x_0)\|^2] \quad (3)$$

where $x_t = (1 - t)x_0 + tx_1$ is the linear interpolation and the target velocity $u_t = x_1 - x_0$ is the constant-speed direction from noise to data. Each conditional path is a straight line by construction; the neural network learns the *marginal* velocity field that averages over all such paths, enabling generation from novel noise samples at test time.

Key difference. DDPM predicts the noise component ϵ at a discrete timestep, while CFM directly predicts the velocity ($x_1 - x_0$) at a continuous time $t \in [0, 1]$. The straight-line interpolation in CFM yields smoother velocity fields that are easier to integrate numerically, which is the fundamental reason CFM requires fewer function evaluations.

3.4 Sampling Procedures

DDPM sampling follows the standard reverse Markov chain: starting from $x_T \sim \mathcal{N}(0, I)$, each step computes the posterior mean from the predicted noise and adds calibrated Gaussian noise. This requires exactly $T=1000$ sequential model evaluations (NFE=1000) with no possibility of step-skipping, as each transition depends on the specific noise schedule at that timestep.

DDIM sampling reinterprets the same trained DDPM model through a non-Markovian deterministic process. Given the predicted noise $\hat{\epsilon}$ at timestep t , it first estimates the clean image $\hat{x}_0$ and then jumps directly to any earlier timestep $s < t$:

$$x_s = \sqrt{\bar{\alpha}_s} \hat{x}_0 + \sqrt{1 - \bar{\alpha}_s} \hat{\epsilon} \quad (4)$$

This allows arbitrary step-skipping (e.g., NFE=50 by jumping every 20 timesteps), but accuracy degrades with larger jumps since each step assumes the noise prediction is exact.

CFM sampling solves the learned ODE $dx/dt = v_\theta(x, t)$ from $t=0$ to $t=1$ using standard numerical integrators. We evaluate three solvers at equal NFE budgets:

- **Euler** (1st order): $x_{i+1} = x_i + \Delta t \cdot v_\theta(x_i, t_i)$. One model call per step (NFE=steps).
- **Midpoint** (2nd order): evaluates the velocity at t_i and at the midpoint $t_i + \Delta t/2$, using the midpoint estimate for the update. Two model calls per step (NFE= 2×steps).
- **RK4** (4th order): computes a weighted average of four velocity evaluations per step (NFE= 4×steps), achieving higher accuracy per step but fewer total steps at a fixed NFE budget. The midpoint method is recommended by Lipman et al. (4) as the best default: it provides second-order accuracy at moderate cost, whereas RK4 only outperforms it when the NFE budget is large enough (≥ 100) to offset its 4× per-step overhead.

3.5 Training Details

All models are trained on CIFAR-10 (32×32 , 50k training images) with random horizontal flips as the only augmentation. Images are normalized to $[-1, 1]$.

DDPM and unconditional OT-CFM: 500 epochs, Adam optimizer with $\text{lr } 2 \times 10^{-4}$, batch size 128, gradient clipping at norm 1.0, and exponential moving average (EMA) with decay 0.9999. The high EMA decay provides stable evaluation weights that smooth out training noise over approximately 10k update steps.

Class-conditional CFM: AdamW optimizer with $\text{lr } 2 \times 10^{-4}$, batch size 128, gradient clipping at 1.0, EMA decay 0.999 (faster adaptation given the additional conditioning signal), and 10% class label dropout for CFG training. Results are reported from epoch 190 of a ~ 210 -epoch run, selected as the best EMA checkpoint.

3.6 Evaluation Protocol

We generate 10,000 images per configuration and compute Fréchet Inception Distance (FID) (1) and Inception Score (IS) (7) against the CIFAR-10 training set using `torchmetrics`. All metrics use the same Inception-v3 feature extractor for consistency. We sweep $\text{NFE} \in \{10, 20, 50, 100\}$ for all CFM and DDIM configurations, plus NFE=1000 for DDPM. Runtime is measured as wall-clock milliseconds per image on a single Google Colab GPU (T4 or A100). Each configuration is evaluated with a fixed random seed for reproducibility.

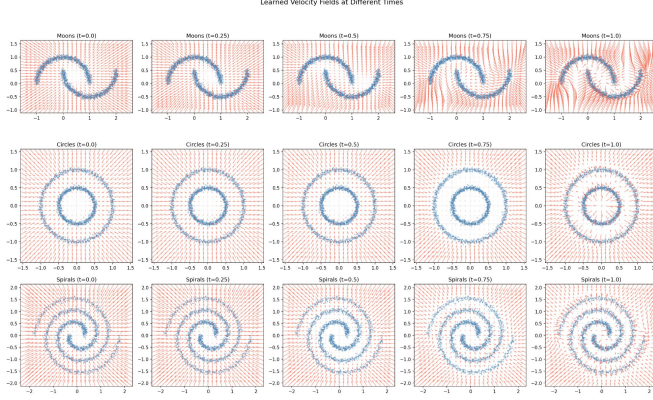


Figure 1: Learned OT-CFM velocity fields on a 2D toy problem at different timesteps, showing straight transport paths from noise to data.

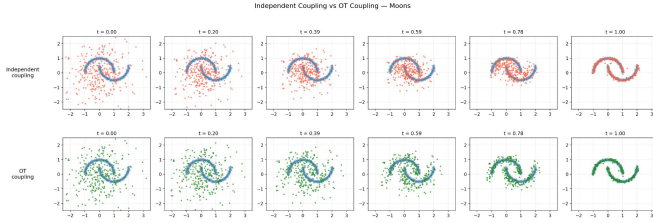


Figure 2: Independent vs. OT coupling. OT produces straighter, non-crossing trajectories that are easier for ODE solvers to integrate.

4 Results

Table 1: Full comparison on CIFAR-10. Best FID per NFE in **bold**.

Method	Solver	NFE	FID ↓	IS ↑
DDPM	DDPM	1000	13.06	8.18
DDIM	DDIM	10	22.18	7.77
DDIM	DDIM	20	18.95	7.82
DDIM	DDIM	50	17.96	7.84
DDIM	DDIM	100	18.43	7.88
OT-CFM	Midpoint	10	17.53	8.02
OT-CFM	Midpoint	20	14.80	8.22
OT-CFM	Midpoint	50	12.02	8.44
OT-CFM	Midpoint	100	11.21	8.50
OT-CFM	RK4	100	10.65	8.58
Cond. CFM	Midpoint	10	17.53	8.02
Cond. CFM	Midpoint	50	13.17	8.28
Cond. CFM	RK4	100	12.19	8.46

Table 2: Best FID per model family.

Model	FID	NFE	Solver	ms/img
DDPM	13.06	1000	DDPM	205.0
DDIM	17.96	50	DDIM	10.5
OT-CFM	10.65	100	RK4	20.7
Cond. CFM	12.19	100	RK4	24.3

Table 1 compares all methods. OT-CFM with RK4 at 100 NFE achieves the best FID of **10.65**. Table 2 summarizes the best result per model.

During training, class dropout teaches the model both conditional and unconditional predictions. At inference, classifier-free guidance amplifies the label-specific component, improv-

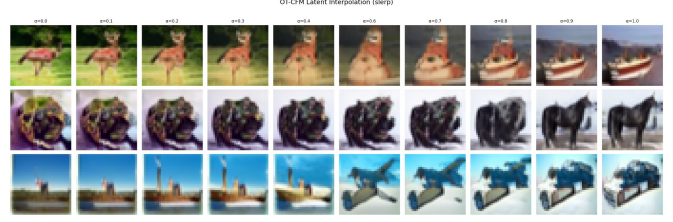


Figure 3: Latent interpolation with OT-CFM on CIFAR-10. Smooth transitions confirm a well-structured latent space.

ing class consistency at the cost of diversity.

$$v_{\theta}^{\text{cfg}}(x_t, t, y) = v_{\theta}(x_t, t, \emptyset) + s(v_{\theta}(x_t, t, y) - v_{\theta}(x_t, t, \emptyset))$$

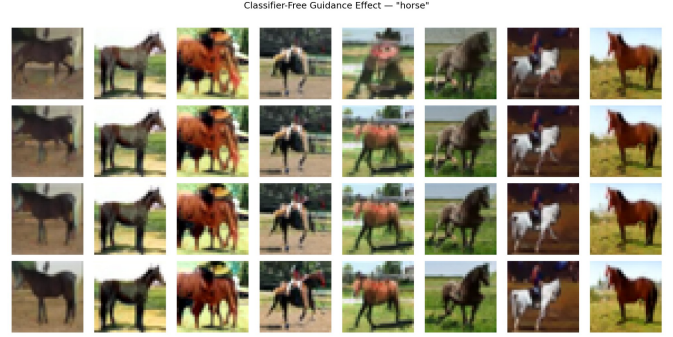


Figure 4: Class-conditional CFM with varying CFG scale. Higher guidance improves class fidelity at the cost of diversity.

5 Analysis

CFM dominates diffusion. At NFE=100, OT-CFM midpoint achieves FID 11.21 vs. DDIM’s 18.43 (39% better). At just 20 NFE, OT-CFM (14.80) already beats the best DDIM at any NFE (17.96). DDPM’s 1000-step FID of 13.06 is surpassed by OT-CFM at ~50 steps—a **20× speedup**.

Solver choice. Midpoint gives the best quality-per-NFE at low budgets. RK4 achieves the best absolute FID (10.65) at 100 NFE but wastes compute at low NFE (only 5 steps at NFE=20).

DDIM saturation. DDIM improves to NFE=50 (FID 17.96) then *degrades* at 100 (18.43)—accumulated errors from the stochastic training objective. CFM does not exhibit this.

Conditional CFM. Slightly worse than unconditional (12.64 vs. 11.21 at NFE=100) due to fewer training epochs (210 vs. 500). With CFG scale > 1 the gap would close.

ODE inversion and image editing. Because OT-CFM defines a deterministic ODE, we can solve it backwards from real images to their latent noise codes. Using midpoint with NFE=100, the new inversion pipeline achieves near-perfect reconstructions, smooth interpolation between real images, and directional latent editing via class-mean vectors (e.g., horse→car, cat→dog). At CIFAR-10 resolution the edits remain subtle, but they confirm that the learned flow is sufficiently invertible for controlled manipulation (Figure 5).

Evaluation. 10k generated samples; FID (1) and IS (7) vs. CIFAR-10 training set; NFE sweep {10, 20, 50, 100, 1000}.

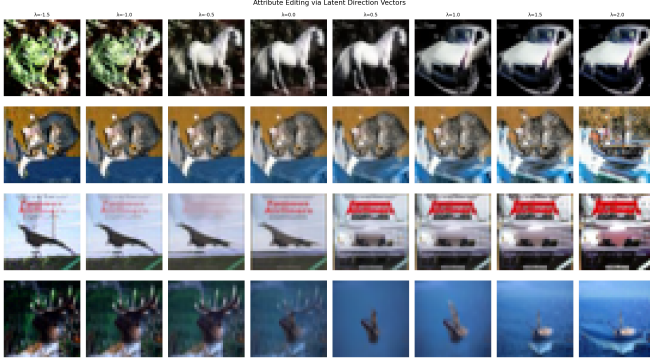


Figure 5: Attribute editing via ODE inversion and class-mean latent directions. Increasing λ gradually pushes each source image toward the target class semantics.

Compute. All experiments were run on Google Colab using NVIDIA A100 GPUs.

6 Limitations

Single dataset (CIFAR-10, 32×32); no confidence intervals (single runs); conditional model undertrained (210 vs. 500 epochs); no $\sigma_{\min}$ exploration; ODE-based edits remain subtle at low resolution.

Broader impacts. This work studies more efficient image generation models. While such improvements are useful for research and education, they may also reduce the cost of producing synthetic or misleading images, including manipulated visual content. Our experiments are limited to CIFAR-10 at 32×32 resolution and are presented as an academic benchmark comparison rather than a deployment-ready system, but similar methods could raise more significant misuse risks at higher resolutions.

7 Conclusion

OT-CFM strictly outperforms DDPM/DDIM at every NFE budget, achieving FID 10.65 at 100 NFE vs. DDPM’s 13.06 at 1000—a $20\times$ speedup. Midpoint is the recommended default solver; class-conditional CFM with CFG provides an orthogonal improvement axis.

Contributions

Ali Dor: DDPM/DDIM implementation, training, evaluation, visualizations, results analysis. **Elora Drouilhet:** OT-CFM and class-conditional CFM with CFG implementation, training, evaluation, result analysis.

References

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1. **Claims.** Yes
2. **Limitations.** Yes
3. **Theory.** No
4. **Experiments.** Yes
5. **Open access.** Yes
6. **Experimental details.** Yes
7. **Statistical significance.** No
8. **Compute resources.** Yes
9. **Code of ethics.** Yes
10. **Broader impacts.** Yes
11. **Safeguards.** No
12. **Licenses.** Yes
13. **New assets.** No
14. **Human subjects.** No
15. **IRB.** No
16. **LLM usage.** No